Name and Section: Points: / 39

MTH:3340 Abstract Algebra Quiz 2
Due: Friday Feb. 6

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: Your ability to express your solutions is just as important as mathematical accuracy. You are expected to write every solution using complete sentences when possible. You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

1. (6 points) Let R be the relation on $\mathbb{Z}$ where nRm only if $nm > 0$ for any integers n and m. Determine whether R is reflexive, symmetric, or transitive. You must check every property.

Reflexive: $0 \in \mathbb{Z}$ and $0 \cdot 0 = 0 \not> 0$ so R is not reflexive.

Symmetric: Let $n, m \in \mathbb{Z}$. If nRm then $n \cdot m > 0$ so $m \cdot n > 0$ thus mRn. Then R is symmetric.

Transitive: Let $n, m, p \in \mathbb{Z}$. If nRm and mRp then $n \cdot m > 0$ and $m \cdot p > 0$. Consider the following cases

If $m > 0$, then $n > 0$ and $p > 0$. Thus $n \cdot p > 0$ and nRp

If $m < 0$, then $n < 0$ and $p < 0$. Thus $n \cdot p > 0$ and nRp.

No integer n, m, p can be zero, so in all cases if nRm and mRp, nRp.

Then R is transitive.

2. (9 points) Let S be the relation on $\mathbb{R}$ where xSy only if $|x-y|$ $\cancel{\leq 4}$ is a multiple of 4 for any real numbers x and y.

(a) Show that S is an equivalence relation.

Reflexive: For any $x \in \mathbb{R}$, $|x-x| = 0 = 4 \cdot 0$ is a multiple of 4, so R is reflexive.

Symmetric. Let $x, y \in \mathbb{R}$, if xRy then $|x-y| = 4k$ for some $k \in \mathbb{Z}_{\geq 0}$. $|x-y| = |-(y-x)| = |y-x|$ so $|y-x| = 4k$ hence yRx and so R is symmetric.

Transitive: Let $x, y, z \in \mathbb{R}$. If xRy and yRz then $|x-y| = 4k$ and $|y-z| = 4l$ for some $k, l \in \mathbb{Z}_{\geq 0}$.

Then $x - y = \pm 4k$ and $y - z = \pm 4l$ so

$$(x-y)+(y-z) = x - z \quad \text{and}$$
$$(x-y)+(y-z) = \pm 4k \pm 4l = \pm 4(k \pm l)$$

so $|x-z| = |\pm 4(k \pm l)|$ is a multiple of 4, thus xRz.

Then R is transitive. Therefore, R is an equivalence relation.

(b) Describe the equivalence classes $[4]_S$ and $[-10]_S$ using set-builder notation.

$[4]_S = \{x \mid x \in \mathbb{R} \text{ and } \underline{|x-4| = 4k} \text{ for some } k \in \mathbb{Z}_{\geq 0}\}$

$x - 4 = \pm 4k$, $x = \pm 4k + 4$ is some multiple of 4

$[4]_S = \{\ldots, -8, -4, 0, 4, 8, \ldots\}$

$[10]_S = \{x \mid x \in \mathbb{R} \text{ and } \underline{|x-10| = 4k} \text{ for some } k \in \mathbb{Z}_{\geq 0}\}$

$x - 10 = \pm 4k$, $x = 10 \pm 4k$ is 10 plus any multiple of 4

$[10]_S = \{\ldots, -2, 2, 6, 10, 14, 18, \ldots\}$

3. (8 points) Prove that if a and b are any integers with the same parity, then $3a+7$ and $7b-4$ do not.

Let a and b be integers. Assume they have the same parity. If a and b are both even then $a=2k$ and $b=2l$ for some integers k, l.

Then $3a+7 = 3(2k)+7 = 2(3k+3)+1$ is odd and

$7b-4 = 7(2l)-4 = 2(7l-2)$ is even so

$3a+7$ and $7b-4$ have opposite parity. If a and b are odd, $a=2k+1$ and $b=2l+1$ for integers k, l instead.

Then $3a+7 = 3(2k+1)+7 = 2(3k+5)$ is even and

$7b-4 = 7(2l+1)-4 = 2(7l+1)+1$ is odd, so they have opposite parity.

Therefore if a and b are integers with equal parity, $3a+7$ and $7b-4$ do not have equal parity ∎

4. (8 points) Prove that for any integer N, if N^2 is not divisible by 4, then N is odd.

Let N be an integer. Assume N is even. Then $N=2k$ for some integer k. Then $N^2 = (2k)^2 = 4k^2$ so N^2 is divisible by 4.

Therefore, by proof by contraposition, if N is an integer and N^2 is not divisible by 4 then N is odd ∎

5. (8 points) Prove that if z is an integer, $5z^2 + 3z + 7$ is odd.

Let z be any integer.

If z is even, then $z = 2k$ for some integer k.

Then $5z^2+3z+7 = 5(2k)^2+3(2k)+7$

$= 2(10k^2+3k+3)+1$ is odd.

If z is odd, then $z = 2l+1$ for some integer l.

Then $5z^2+3z+7 = 5(2l+1)^2+3(2l+1)+7$

$= 5(4l^2+4l+1)+6l+3+7$

$= 20l^2+26l+15$

$= 2(10l^2+13l+7)+1$ is odd.

In either case, $5z^2+3z+7$ is odd.

Therefore, if z is any integer, $5z^2+3z+7$ is odd. ☒